

Full Stack Development with MERN

Project Documentation Format

Project Title: ServiceNow-Based Employee Laptop Request Application

Platform Note: The supplied FSD template is written for a MERN application. This project is implemented on ServiceNow, so the same documentation sections are mapped to ServiceNow equivalents rather than inventing React/Node/MongoDB components.

1. Introduction

Project Title: ServiceNow-Based Employee Laptop Request Application

Team Members: Team member names and individual roles were not provided in the source project material. The project records show the work assigned to Gurram Prasanna Sai.

2. Project Overview

The project implements a centralized and controlled process for managing employee laptop requests in an IT organization. It replaces manual email and verbal requests with a standardized digital workflow.

- Employees submit a Laptop Request through a ServiceNow Record Producer.
- Requests distinguish Basic and Advance laptop usage.
- A Business Rule prevents duplicate laptop requests within the same calendar year.
- Flow Designer starts manager approval automatically and updates the request status.
- Approval/rejection notifications keep the requester informed.
- Roles, groups and ACLs control employee and manager access.

3. Architecture

Frontend: Service Portal / Service Catalog Record Producer provides the employee-facing request form.

Backend: ServiceNow Business Rules provide validation and policy enforcement, while Flow Designer provides the approval workflow and notifications.

Database: A custom *Laptop Request Form* table stores request data, with reference fields linked to the User table. The approval relationship connects requests with *sysapproval_approver*.

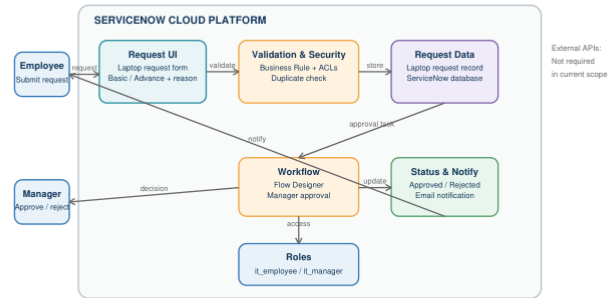
Project Design Phase Solution Architecture

Date	15 February 2025
Team ID	
Project Name	ServiceNow-Based Employee Laptop Request Application
Maximum Marks	4 Marks

Solution Architecture:

The architecture bridges the employee laptop-request business problem and the ServiceNow technology solution. It centralizes the request, applies validation and access controls, routes approval and stores the request and status on the ServiceNow platform.

Solution Architecture Diagram:



Architecture Flow: Employee → Request Form → Validation & ACLs → ServiceNow Request Record → Flow Designer → Manager Approval → Status Update & Notification.

Key technologies: ServiceNow UI / Record Producer, Business Rules, Flow Designer, ServiceNow database, roles and ACLs, and ServiceNow Cloud Platform.

Figure 1. Solution architecture prepared from the project design artifact.

4. Setup Instructions

Prerequisites: A ServiceNow development instance with suitable application-development permissions. The project screenshots also show use of an elevated role when creating ACLs.

Implementation sequence:

- Create/select an Update Set and make it current.
- Create the custom application and define its scope/access settings.
- Create the Laptop Request Form table and configure its fields.
- Create the Employee Service Catalog entry and Laptop Request Record Producer.
- Configure the approval relationship with sysapproval_approver.
- Create the Business Rule for yearly duplicate-request prevention.
- Create IT Manager and IT Employee roles and groups.
- Configure table-level ACLs for read/create/write access.
- Build the Flow Designer approval workflow and email notifications.
- Test the request lifecycle and commit/export the Update Set for deployment.

5. Folder Structure

The supplied template expects Client and Server folders. This ServiceNow project does not use a local MERN client/server folder structure. Its logical structure is organized by ServiceNow application artifacts:

- Application: Project Application
- Data: Laptop Request Form table and configured fields
- Catalog: Employee Service Catalog and Laptop Request Record Producer
- Logic: Business Rules
- Workflow: Flow Designer approval flow
- Security: Roles, Groups and ACLs
- Deployment: Update Set

6. Running the Application

The project is executed inside the configured ServiceNow instance rather than with local *npm start* commands. Employees access the Laptop Request entry through the Service Catalog/Service Portal.

Typical request lifecycle: **Record Created** → **Pending Approval** → **Manager Approval** → **Approved/Rejected** → **Notification**.

7. API Documentation

No custom REST API endpoints are documented in the supplied project material. The implementation relies on native ServiceNow platform operations, table records, Record Producer mapping, Business Rules and Flow Designer.

8. Authentication

Authentication is handled by the ServiceNow platform. Authorization is implemented through the project-defined roles, groups and ACLs.

Role	Purpose / Access
it_employee	Create and manage laptop requests; read request status.
it_manager	Read-only access to laptop request records for review/approval.

9. User Interface

The project uses a ServiceNow Service Portal/Service Catalog interface. The Record Producer provides the employee-facing Laptop Request form.

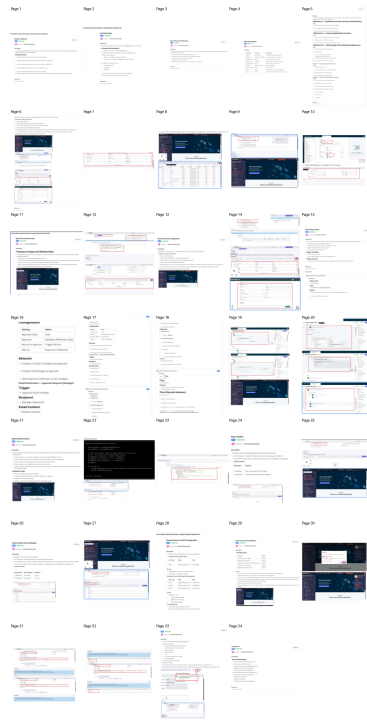


Figure 2. Project implementation screenshots; the source project PDF contains the detailed configuration screens.

10. Testing

Testing covered request creation, field/choice validation, duplicate-request prevention, role-based access and the approval workflow. The project records specifically validate that an employee can create/edit a request, a second request in the same year is blocked, and managers can view but not create/modify requests.

Scenario	Expected Result
Employee creates request	Allowed
Employee edits request	Allowed
Employee submits second request in same year	Blocked
Manager views request list	Allowed
Manager tries to edit/create	Blocked

11. Screenshots or Demo

The supplied project PDF contains configuration and validation screenshots across the implementation, including Record Producer, Flow Designer, Business Rule, roles/groups, ACLs and update-set deployment.

12. Known Issues

- No custom API or external integration is documented in the supplied material.
- Actual production deployment details are not specified; the project documents focus on development-instance configuration and Update Set deployment.
- The source template asks for a MERN folder structure, but the actual solution is ServiceNow-based.

13. Future Enhancements

- Extend approval workflows and notifications.
- Add reporting and dashboards for IT and management.
- Integrate laptop provisioning and asset-management activities after approval.
- Add richer request tracking and operational metrics.
- Introduce additional automation around fulfillment.